AudioReach Signal Processing Framework Overview

Qualcomm Technologies, Inc.

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

80-VN500-3 Rev. C

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets

NO PUBLIC DISCLOSURE PERMITTED: Please report postings of this document on public servers or websites to: DocCtrlAgent@qualcomm.com.

Restricted Distribution: Not to be distributed to anyone who is not an employee of either Qualcomm Technologies, Inc. or its affiliated companies without the express approval of Qualcomm Configuration Management. Distribution to anyone who is not an employee of either Qualcomm Incorporated or its affiliated companies is subject to applicable confidentiality agreements.

Confidential and Proprietary – Qualcomm Technologies, Inc.

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
jijiasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets

NO PUBLIC DISCLOSURE PERMITTED: Please report postings of this document on public servers or websites to: DocCtrlAgent@qualcomm.com.

Restricted Distribution: Not to be distributed to anyone who is not an employee of either Qualcomm Technologies, Inc. or its affiliated companies without the express approval of Qualcomm Configuration Management. Distribution to anyone who is not an employee of either Qualcomm Incorporated or its affiliated companies is subject to applicable confidentiality agreements.

Not to be used, copied, reproduced, or modified in whole or in part, nor its contents revealed in any manner to others without the express written permission of Qualcomm Technologies, Inc.

All Qualcomm products mentioned herein are products of Qualcomm Technologies, Inc. and/or its subsidiaries.

Qualcomm is a trademark of Qualcomm Incorporated, registered in the United States and other countries. Other product and brand names may be trademarks or registered trademarks of their respective owners.

This technical data may be subject to U.S. and international export, re-export, or transfer (“export”) laws. Diversion contrary to U.S. and international law is strictly prohibited.

Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

©2019-2020 Qualcomm Technologies, Inc. and/or its subsidiaries. All rights reserved.

Revision History

Revision	Date	Description
A	October 2019	Initial release
B	November 2019	Editorial update
C	September 2020	Terminology updates for AudioReach

Note: There is no Rev. I, O, Q, S, X, or Z per Mil. standards.

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

Contents

- Introduction
- Building blocks
- Concepts
- Advantages
- Workflow
- References
- Questions?

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

Introduction

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

Introducing the AudioReach™ Signal Processing Framework

- New signal processing framework that provides the means to setup, configure, and execute signal processing modules in various use cases

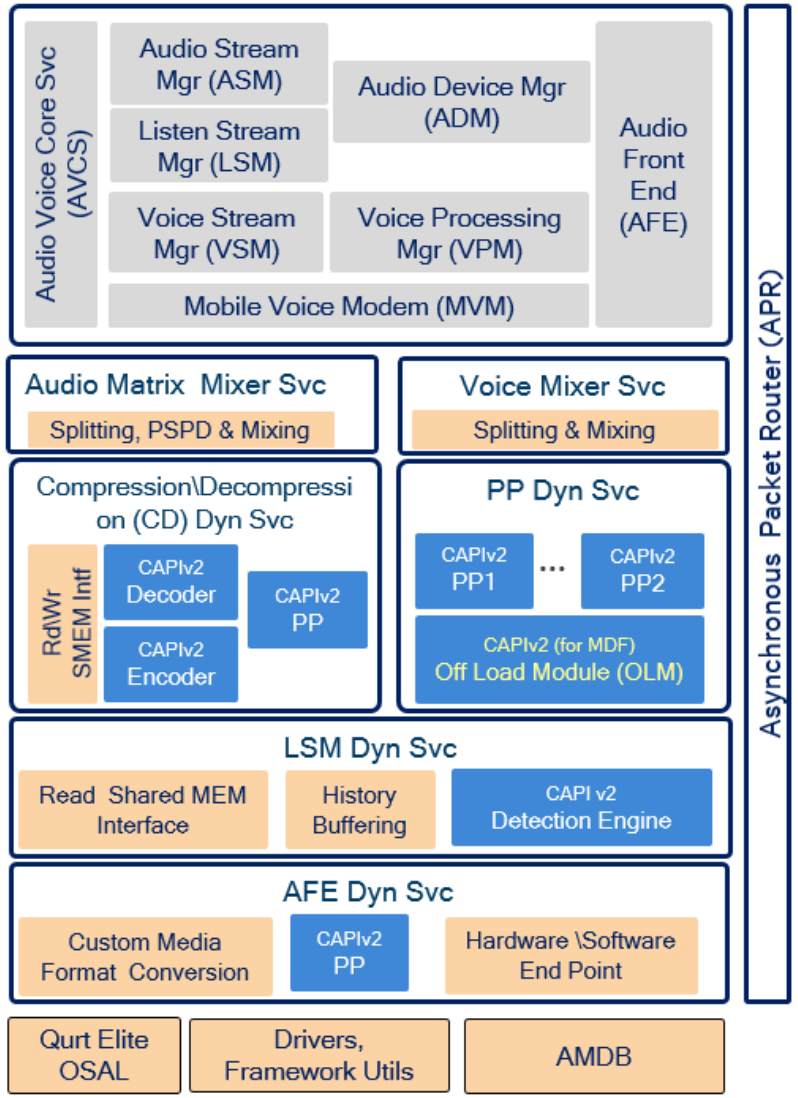
- Replaces the Elite framework
  - Introduced in 2011 on the Qualcomm Hexagon™ DSP
  - Evolved significantly to meet various requirements
    - From simple playback, capture, voice call to advanced Bluetooth, voice UI, and LPI use cases
    - Support for various latency and power modes
    - Module customizations from APPI and CAPI to CAPI v2
    - Real-time calibration (RTC) and real-time monitoring (RTM)
    - Module discovery with h2xml

- What makes the AudioReach™ signal processing framework (SPF) more competitive?
  - SPF offers support for:
    - Non-linear graphs and flexible data routing
    - Generalized multiport modules
    - Metadata propagation and floating-point data format
    - Real time graph management (RTGM)
  - Framework customization
  - Aggressive memory scalability across the products

Qualcomm Confidential – May Contain Trade Secrets 2022-11-05 09:29:49 GMT hasheng@longcheer.com

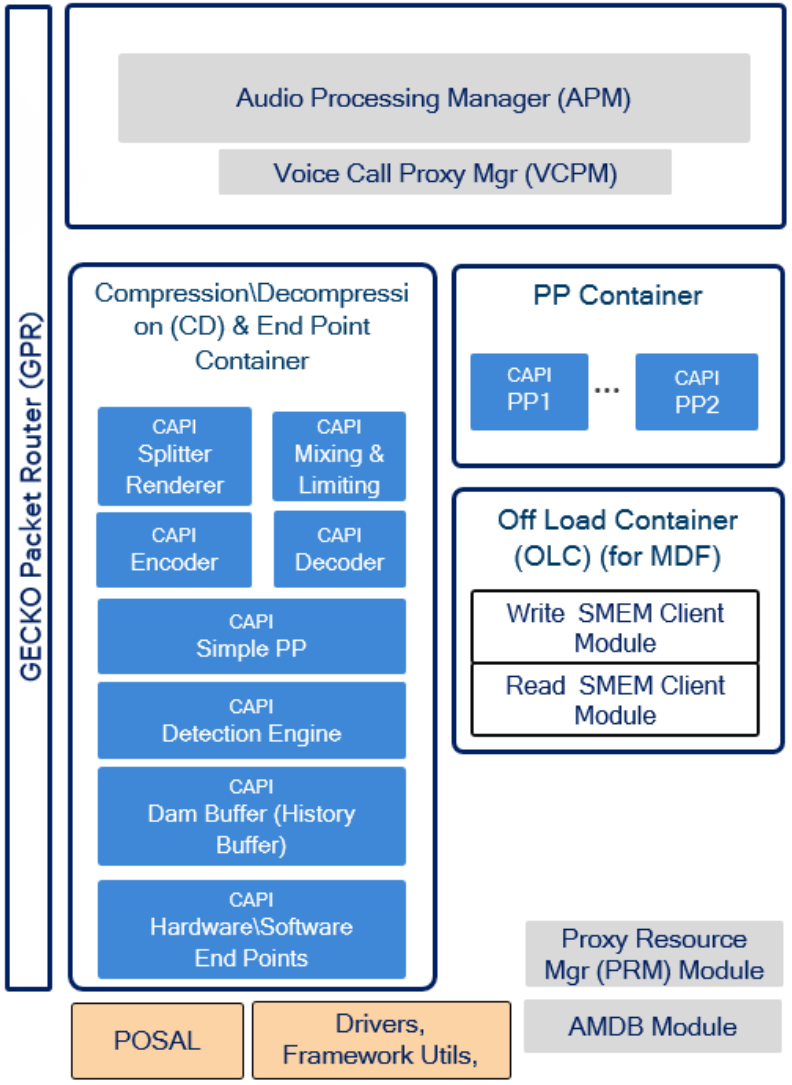
Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Migrating from Elite to SPF



Elite -> SPF

- Service and port-based addressing (APR) to unique instance ID addressing (GPR)
- Unified graph management – APM
- Proxy manager – VCPM handles voice call special needs
- Fully modular and use case agnostic
- Linear topologies to generalized graphs (DAG only, no feedback loops inside SPF)
- Embed (real time, low power, low latency, LPI, etc.) & Open access friendly
- Platform agnostic (no longer a Hexagon (Q6) only framework)
- Multi-core and multi-instance capable
- Additional capabilities such as multi-port modules, metadata, generalized IMC, etc.



Not customizable | Static Svc\Modules | Customizable

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Building Blocks

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

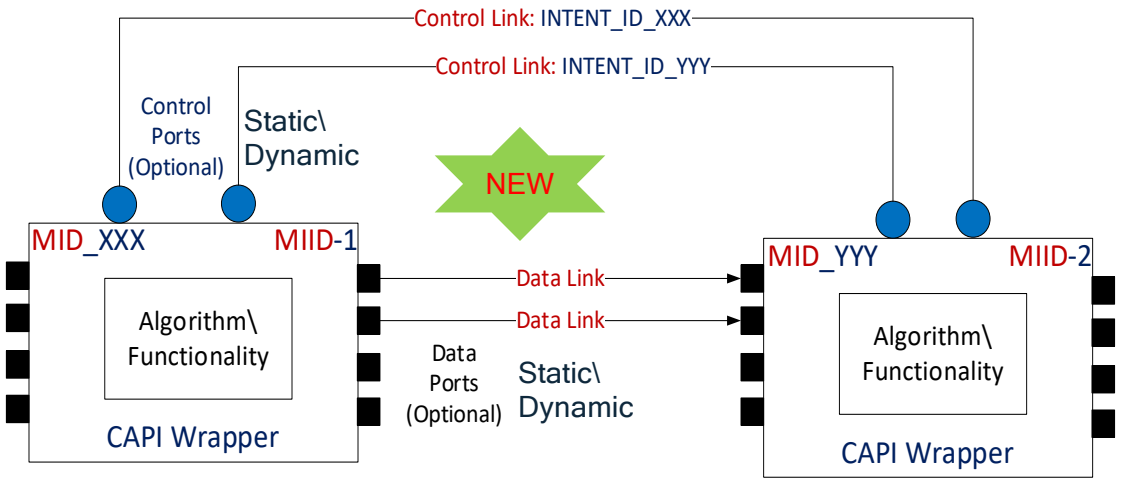
Modules and Links

- Module
  - Addressable functionality in SPF
- Module ID (MID)
  - Unique ID to identify the module
  - Useful during graph creation and shared object identification in dynamic download
- Module instance ID (MIID)
  - Unique ID to identify the instance of module
  - Created in the QACT™ tool at graph design time for dynamic instance IDs
  - Used as “address” to access a module
- Control modules
  - Static framework modules that provide public interfaces to manage the resources in the SPF
- Data-path modules
  - Provide defined functionality in the data path
  - Wrapped with Common Audio Processing Interface (CAPI)
  - Modules can be static or dynamic
- Control link
  - Bi-directional, point-to-point, dynamic/static
  - Used for exchanging control messages (that do not need to be synchronous with data) between two data path modules
- Data link
  - Uni-directional, point-to-point, dynamic/static
  - Used for exchanging data messages (that need to be synchronous with data buffers\samples) between two data path modules

Control Modules (Static)

APM APM_MODULE_INSTANCE_ID – 0x1	PRM PRM_MODULE_INSTANCE_ID – 0x2
AMDB AMDB_MODULE_INSTANCE_ID – 0x3	VCPM VCPM_MODULE_INSTANCE_ID – 0x4

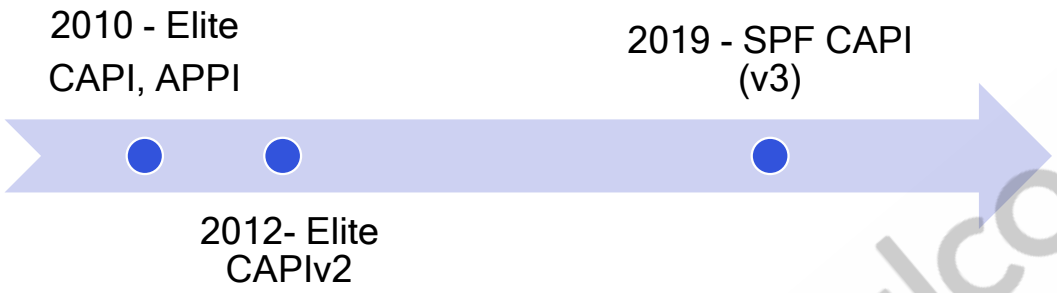
Data-path Modules (Static\Dynamic)



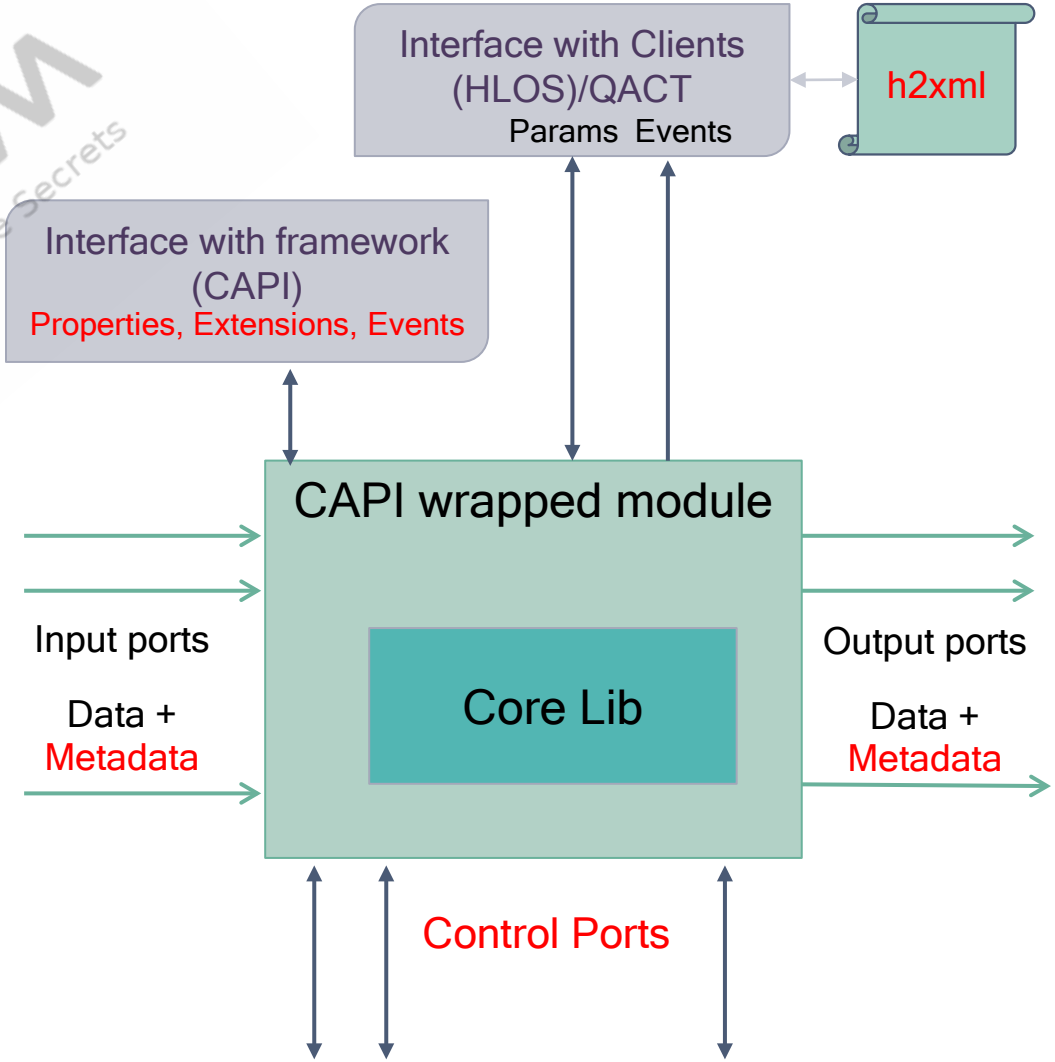
Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longsheer.com

Common Audio Processing Interface (CAPI) v3

- History



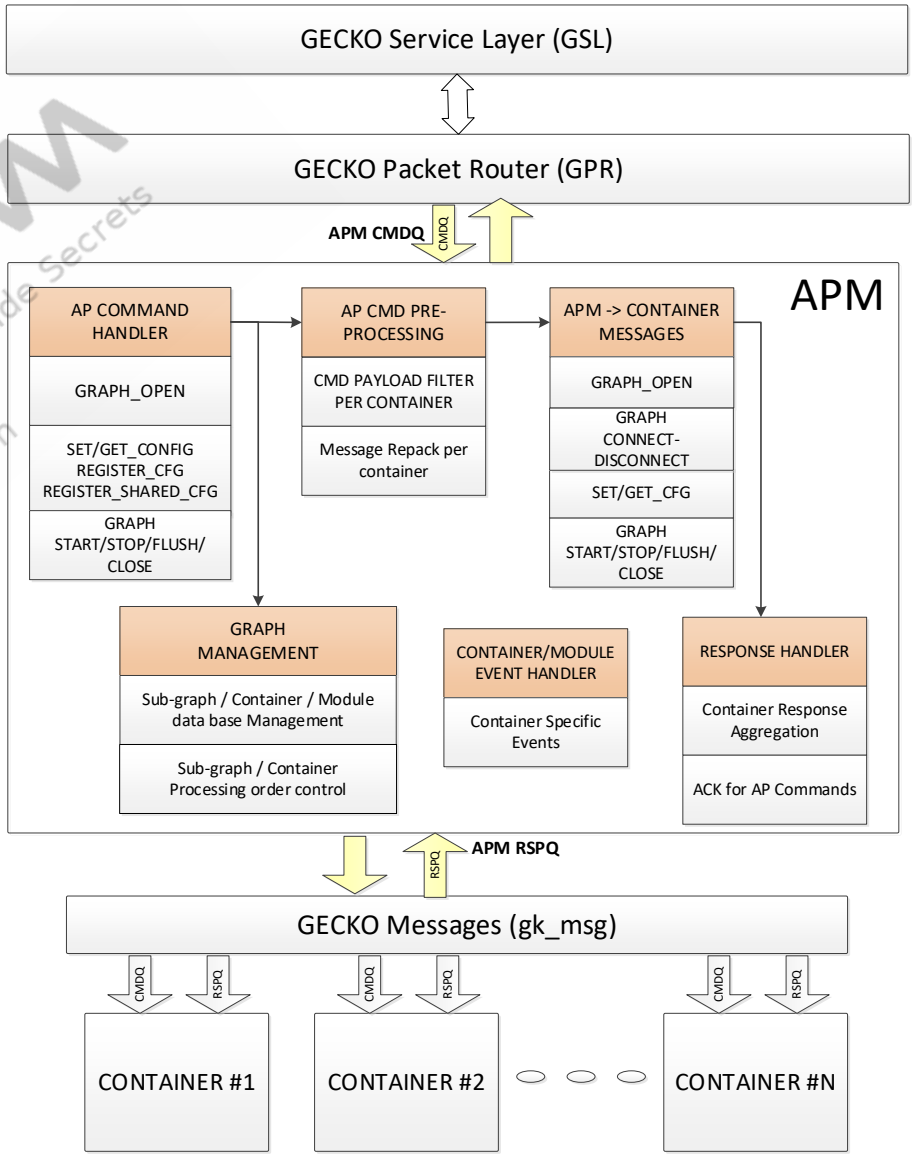
- Version #3 enables the support for
  - Generalized multi-port (data and control) modules
  - Meta data propagation
  - Awareness to data flow states
  - Additional capabilities through h2xml discovery method
- Migration (from CAPIv2 to CAPI version #3)
  - Renamed simply as CAPI and versioned as v3. Mostly backward compatible with CAPIv2.
  - Most modules can be modified with a script to be compatible. The script also outputs if any manual changes are needed.
  - Modules like echo cancellation (EC), detection engines, and encoders need manual changes. Examples will be given along with close support.
  - H2XML annotation needs to be re-done. Mapping script/doc to be provided.
  - All modules must be re-tested on SPF



Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

Audio Processing Manager (APM)

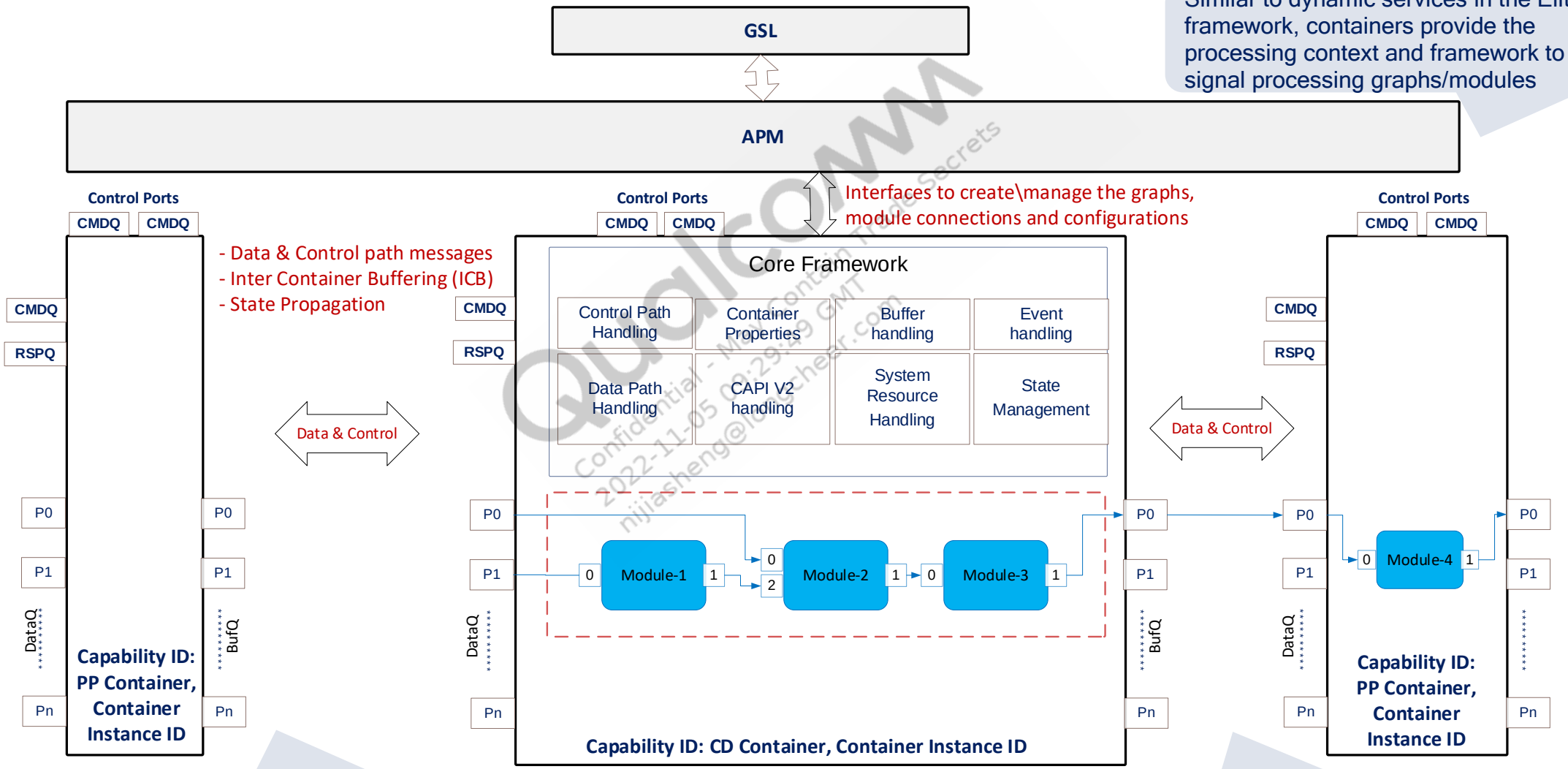
- Static module that interacts with the client and manages all graphs in the SPF
- Interfaces with GSL via GPR via public interfaces, such as:
  - APM_CMD_GRAPH_OPEN, APM_CMD_GRAPH_CLOSE
  - APM_CMD_SET_CFG, APM_CMD_GET_CFG
  - APM_CMD_GRAPH_PREPARE,
  - APM_CMD_GRAPH_START, APM_CMD_GRAPH_STOP
- Works closely with containers to:
  - Create and configure container processing threads
  - Set up a module's graph within a container
  - Provide the module configuration/calibration
  - Manage the data/control path connections across containers
  - Distribute graph-specific operations across the respective containers
  - Manages the necessary sequences to start or stop the graphs across containers
- Provides shared memory mapping, path delay query, and global reset functionalities



Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

Containers

Similar to dynamic services in the Elite framework, containers provide the processing context and framework to the signal processing graphs/modules



Methods to exchange messages between containers is same as in the Elite framework

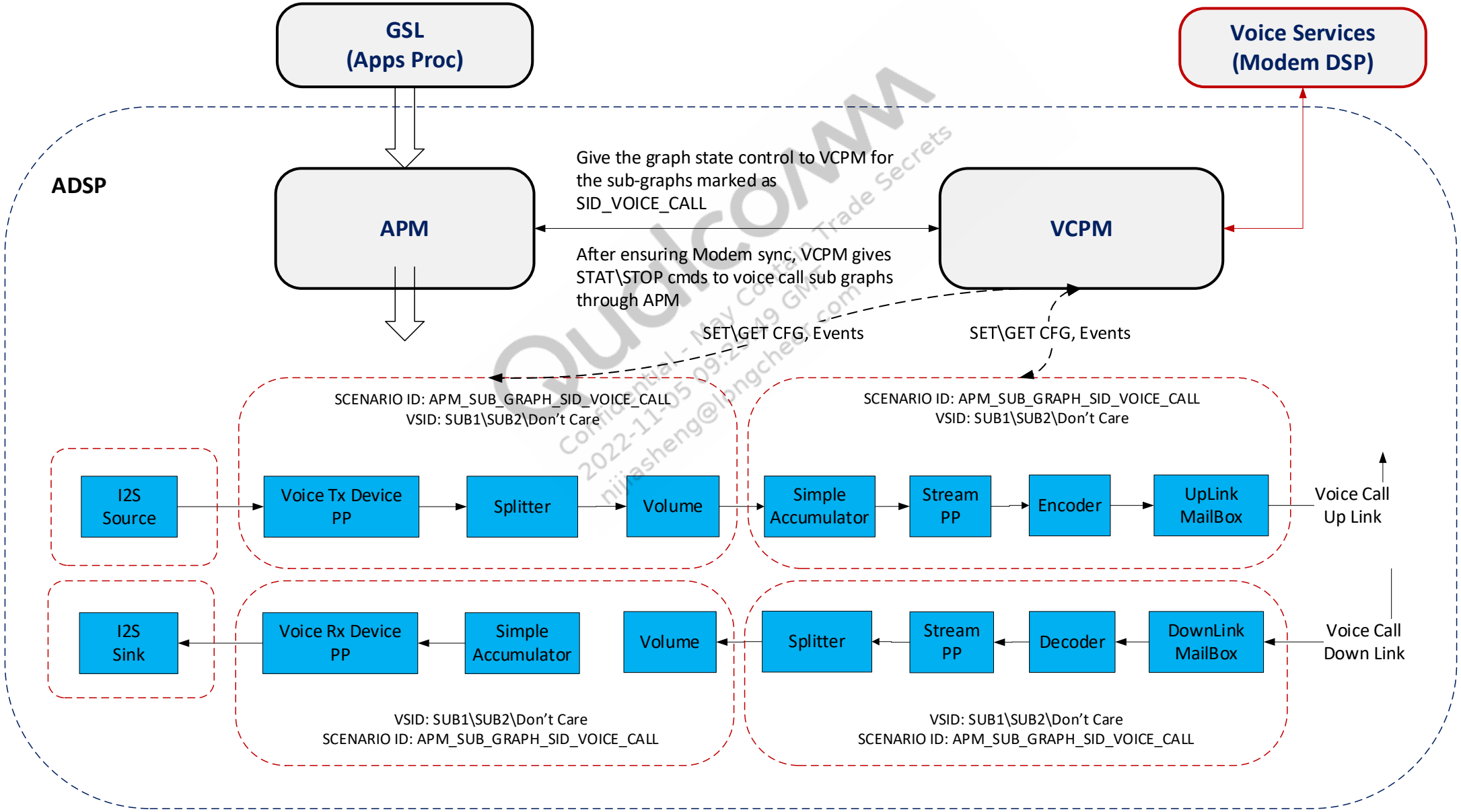
Custom containers are allowed in SPF and can be inter-operable

NEW

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@lobocheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Voice Call Proxy Manager (VCPM)



Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

VCPM (cont.)

- VCPM is a helper module to APM that handles voice call-specific needs, including:
  - Handling synchronization with modem layers during setup, tear-down, and handovers
  - Handling timing offset changes
  - Device switches
  - Floor clock voting for voice-specific graphs to manage the RTD
  - Multi-dimensional calibration
  - Managing dual voice call sessions using APM_SUB_GRAPH_VSID_SUB1 and APM_SUB_GRAPH_VSID_SUB2

S. NO	Elite - Keys	SPF Keys
1	Network ID	VCPM_CAL_KEY_ID_NETWORK_ID
2	Tx Sampling Rate	VCPM_CAL_KEY_ID_SAMPLING_RATE
3	Rx Sampling Rate	
4	Tx Operating mode	
5	Rx Operating mode	
6	Vocoder type	VCPM_CAL_KEY_ID_VOCODER_CLASS
7	FEATURE ID	VCPM_CAL_KEY_ID_BWE
8	Volume Level	VCPM_CAL_KEY_ID_VOLUME_LEVEL
9	Volume Boost	VCPM_CAL_KEY_ID_VOLUME_BOOST

NEW
- All keys in SPF are optional (unlike in the Elite framework where all were mandated)

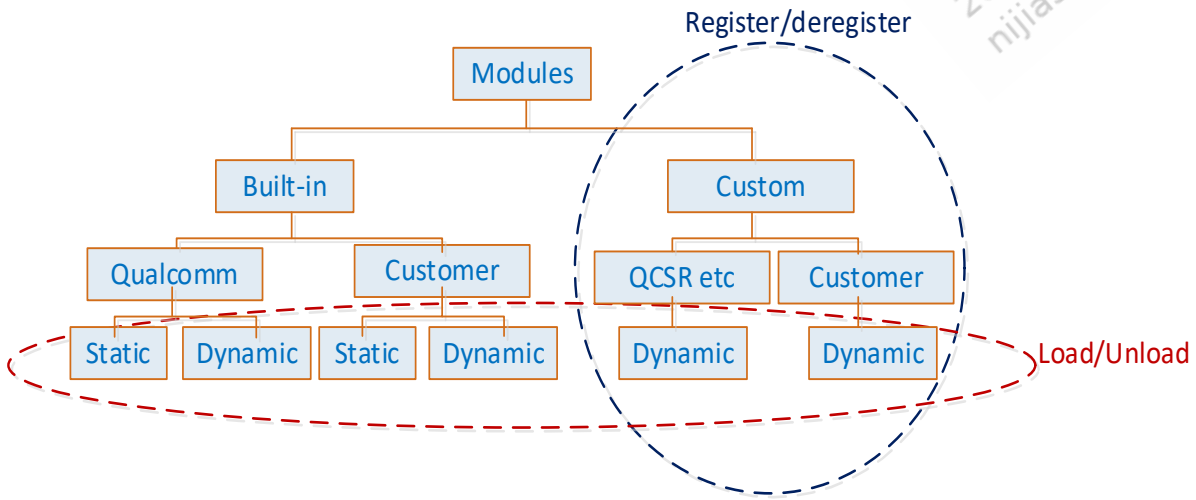
NEW
- SPF design allows for the definition of custom keys without code changes in VCPM
- Compressed blob support is present (similar to in the Elite framework) to optimize calibration memory usage

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@lonacher.com

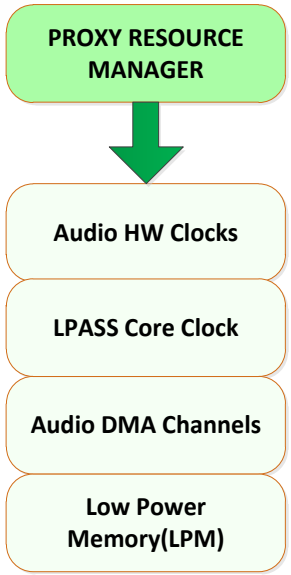
Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

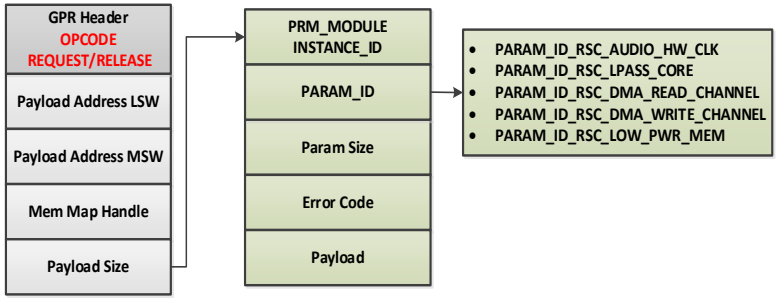
AMDB and PRM

- Audio module database (AMDB)
  - Majorly leveraged from the Elite framework
  - GSL interacts with AMDB for custom modules through:
    - AMDB_CMD_REGISTER_MODULES
    - AMDB_CMD_DEREGISTER_MODULES,
    - AMDB_CMD_LOAD_MODULES
    - AMDB_CMD_UNLOAD_MODULES
  - Supports offload framework
  - adsp_avs_config.acdb
    - The Elite framework read this file directly to learn about pre-load and use case time loading
    - In SPF, this information is merged into a regular ACDB file. GSL provides load/unload commands at boot time or use case setup time.



- Proxy Resource Manager (PRM) – Used to manage and share hardware resources from the underlying platform to the clients





Qualcomm Confidential - May Contain Trade Secrets 2022-11-07 09:29:49 GMT nijiasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

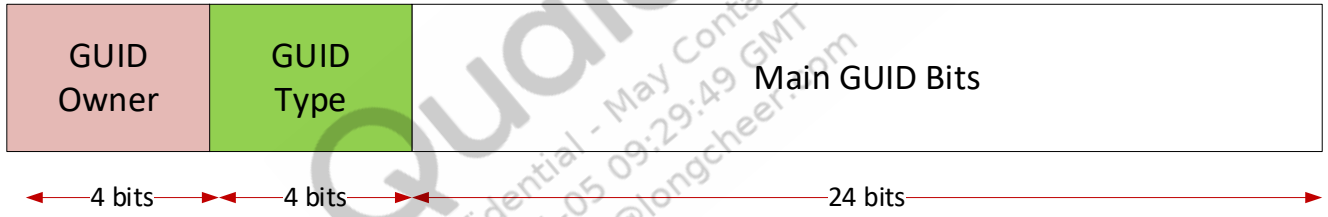
Concepts

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

GUID Format

- SPF follows the format shown when defining GUIDs for various needs
- This format helps the SPF packet router (GPR) handle events/responses, as well as assisting in debugging
- Customers can define module IDs (mandated due to global scope) and parameter IDs (optional due local scope with module) in the non-Qualcomm range

NEW

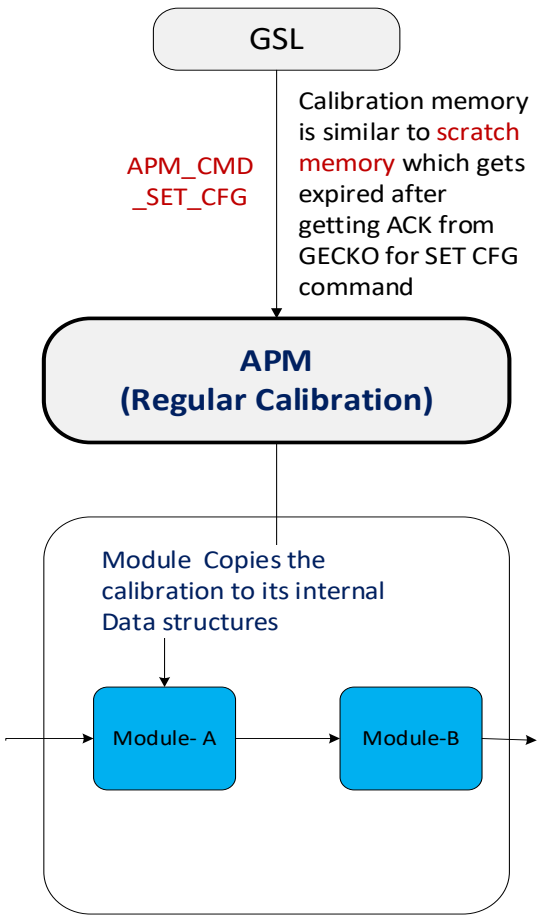


- MM_GUID_OWNER_QC 0x0
- MM_GUID_OWNER_NON_QC 0x1

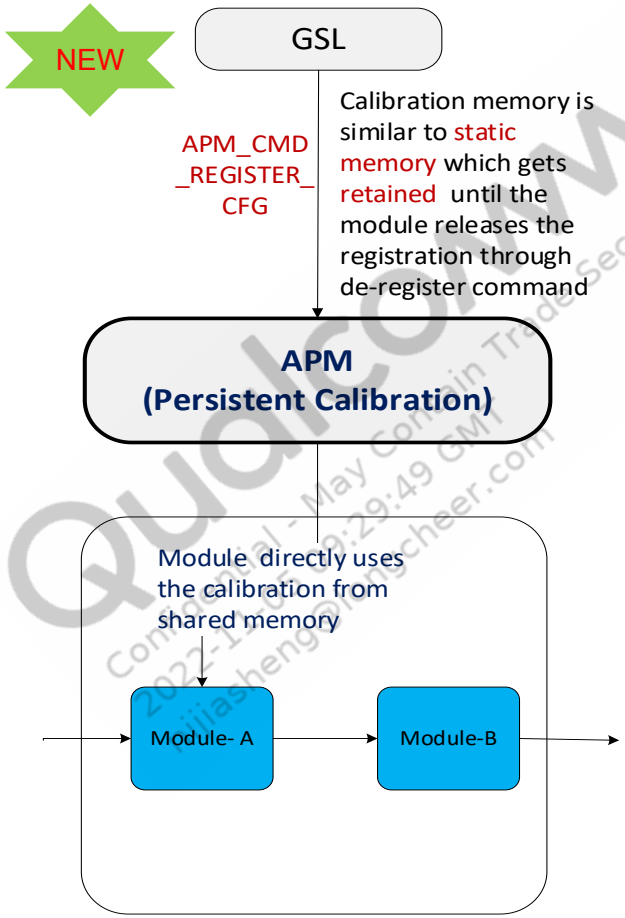
- MM_GUID_TYPE_RESERVED 0x0
- MM_GUID_TYPE_CONTROL_CMD 0x1
- MM_GUID_TYPE_CONTROL_CMD_RSP 0x2
- MM_GUID_TYPE_CONTROL_EVENT 0x3
- MM_GUID_TYPE_DATA_CMD 0x4
- MM_GUID_TYPE_DATA_CMD_RSP 0x5
- MM_GUID_TYPE_DATA_EVENT 0x6
- MM_GUID_TYPE_MODULE_ID 0x7
- MM_GUID_TYPE_PARAM_EVENT_ID 0x8
- MM_GUID_TYPE_FORMAT_ID 0x9
- MM_GUID_TYPE_CAPI 0xA

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nlyasheng@longcheer.com

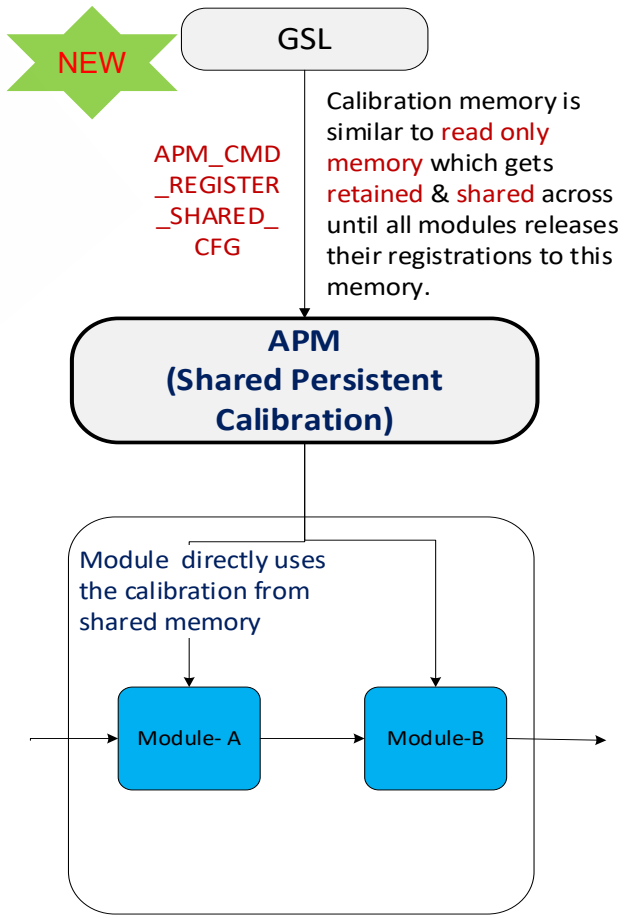
Calibration



- **Useful for small calibration blobs**



- **Useful for large calibration blobs (Read Only or Read-Write).**
- **Avoids additional memory and copy overheads**
- **Used with one module instance only**



- **Useful for large calibration blobs (Read Only).**
- **Avoids additional memory and copy overheads**
- **Multiple module instances can share the same calibration**

Machine Learning (ML) Model Friendly

Qualcomm Confidential - May Contain Trade Secrets 2022-12-09 09:29:49 GMT jijiasheng@longcheer.com

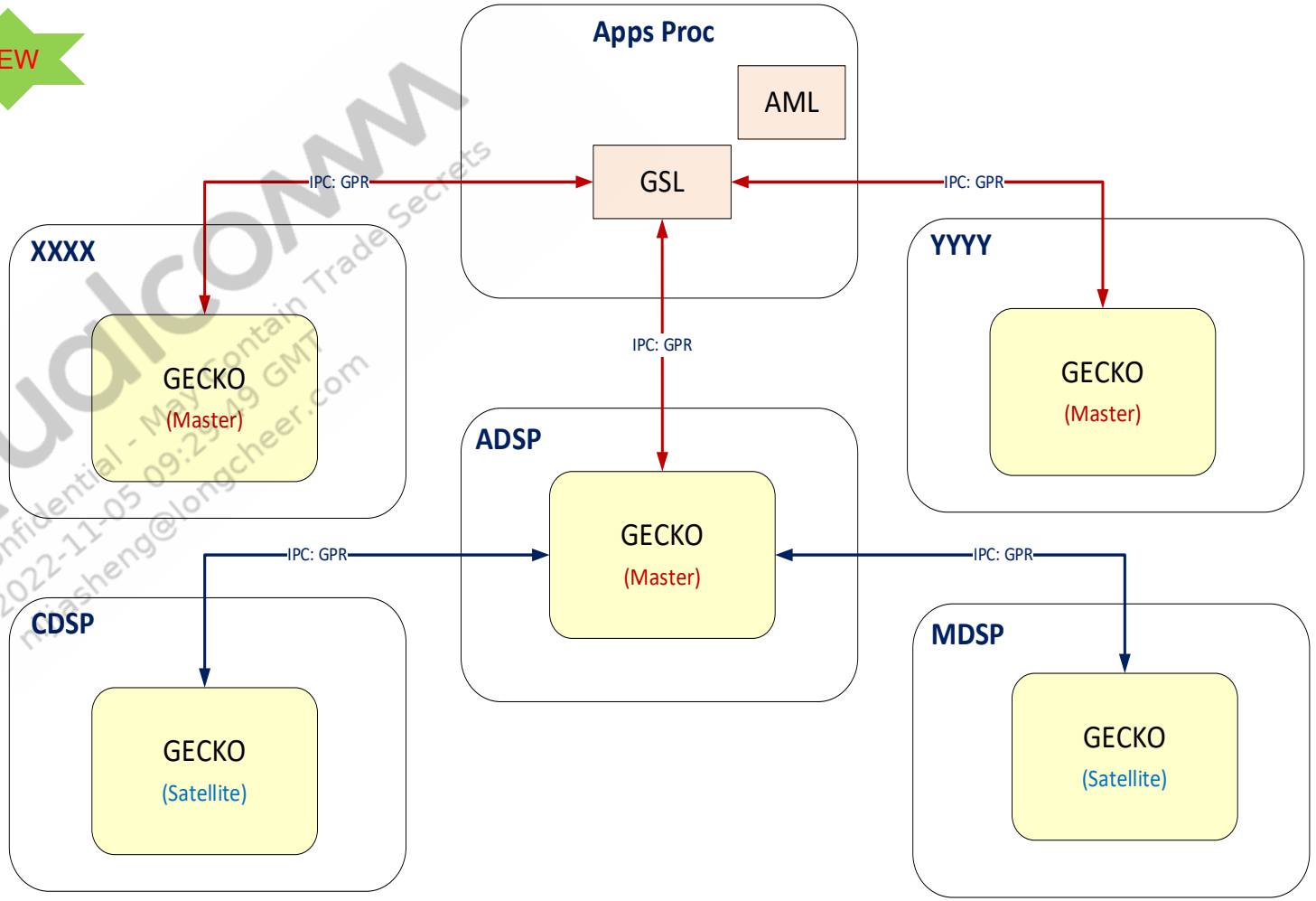
Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Multi-Instance and Multi-Core

- GSL uses the routing ID to know which process domain SPF master can execute the graph
- APM (in the SPF master) detects the offloaded part of the graph and manages through the offload container (OLC)
- Subsystem restart (SSR) and process domain restart (PDR)
  - If the master crashes, restart all use cases associated with that master SPF including any satellite restarts
  - If a satellite crashes, restart all use cases associated with that satellite
  - If GSL crashes, reset all SPF masters and satellites

NEW

NEW



Enhanced SSR/PDR handling

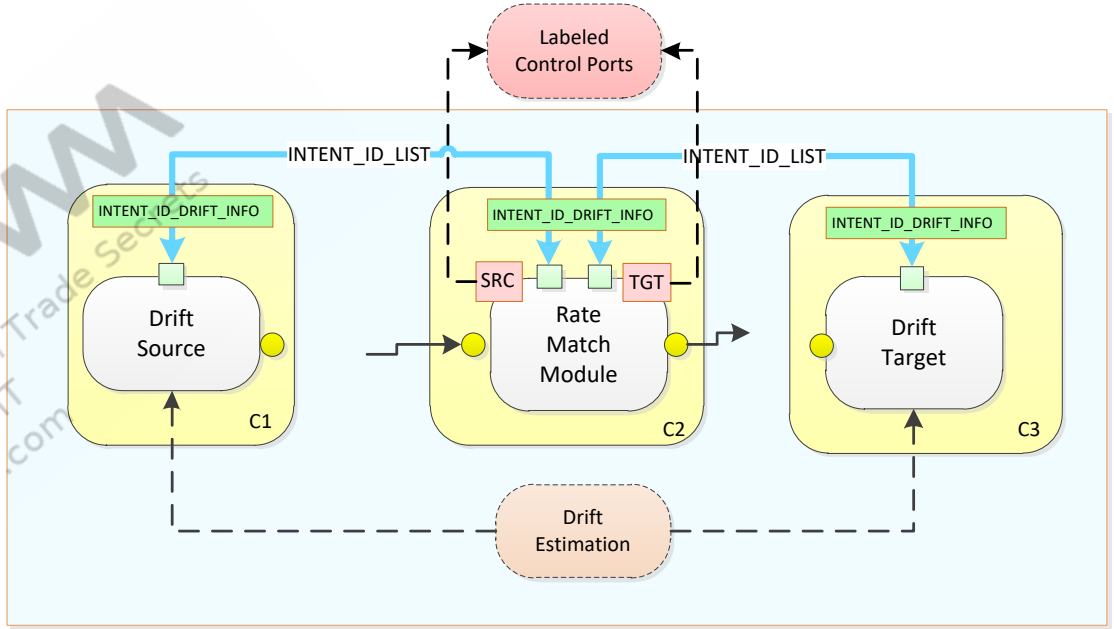
Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:25:49 GMT niasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Rate Matching

- Control links are established between the drift estimator and the rate matching (RM) module
- Drift estimators (DE)
  - Publishes “dynamic” control ports for sharing drift information to the RM module
  - Depending upon the position of the RM module relative to the DEs, it can act as either a source or target for the RM module
- RM module
  - Publishes static control ports receiving drift information
  - Static ports are labeled as source and target
  - Source implies the data rate to match from
  - Target implies the data rate to match to
- Elite vs. SPF
  - Elite framework – RM modules and their positions are fixed and not customizable
  - SPF – Flexibility to customize RM modules and their positions per system designer needs

NEW



New use case enablement is easy

Qualcomm Confidential – May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Other Concepts

- Generalized event framework
  - Allows clients to register for events with a module through APM_CMD_REGISTER_MODULE_EVENTS
  - Provides event-specific configuration during registration
- Metadata propagation
  - Framework propagates generic metadata in the data path
- Path delay
  - Framework allows the clients to read the path delay between two modules (source module port to destination module port)
  - Example use cases – AV sync, path delay for voice call to compute system resources, etc.
- Trigger policies
  - Framework handles the default trigger policies needed for the majority of use cases and modules
  - Framework also provides flexibility for allowing modules to influence trigger policy decisions. This is helpful in complex modules like the Audio Dam module, which handles FTRT and RT data.
- Resource voting
  - Framework votes for higher system resources (clock and bandwidth) during the use case setup and tear-down to reduce latencies
  - All system resource voting requests goes though the power management (PM) server (abstracts underlying platform specific resource management software layers like ADSPPM, sleep drivers, etc.)

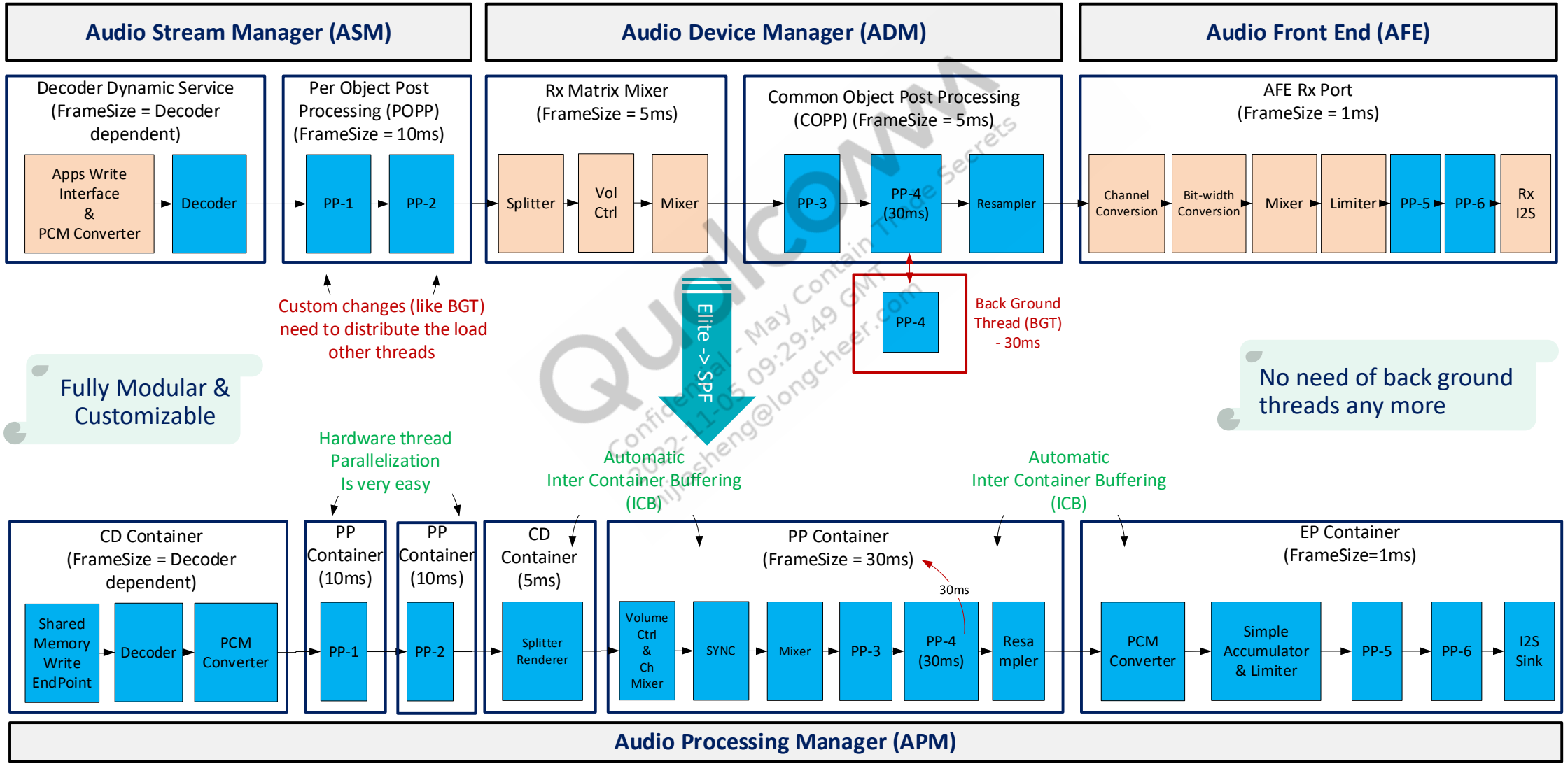
Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 06:29:49 GMT jiajsheng@longsheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Advantages

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

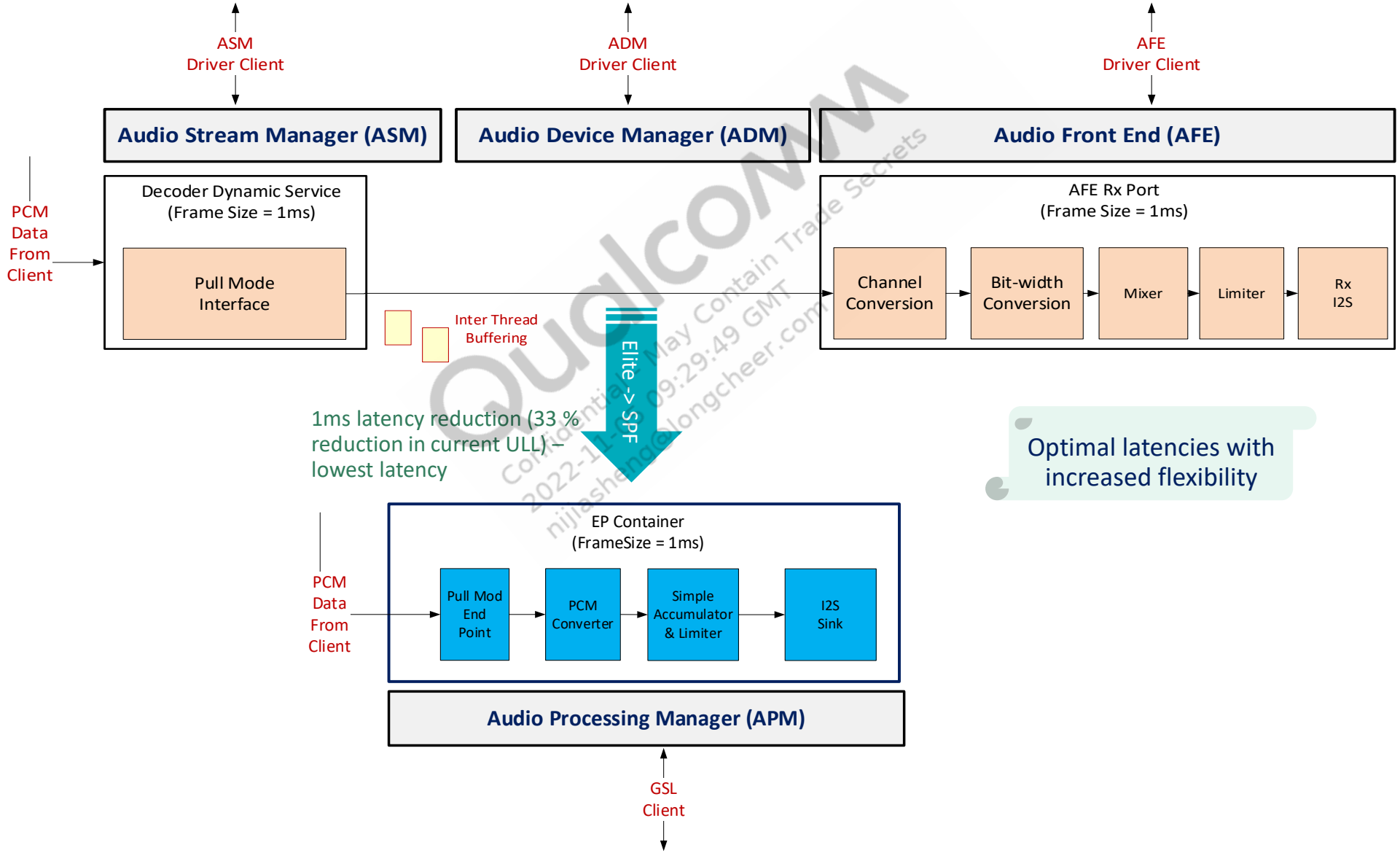
Audio Playback



Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT zhijyesheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Ultra Low Latency Audio Playback

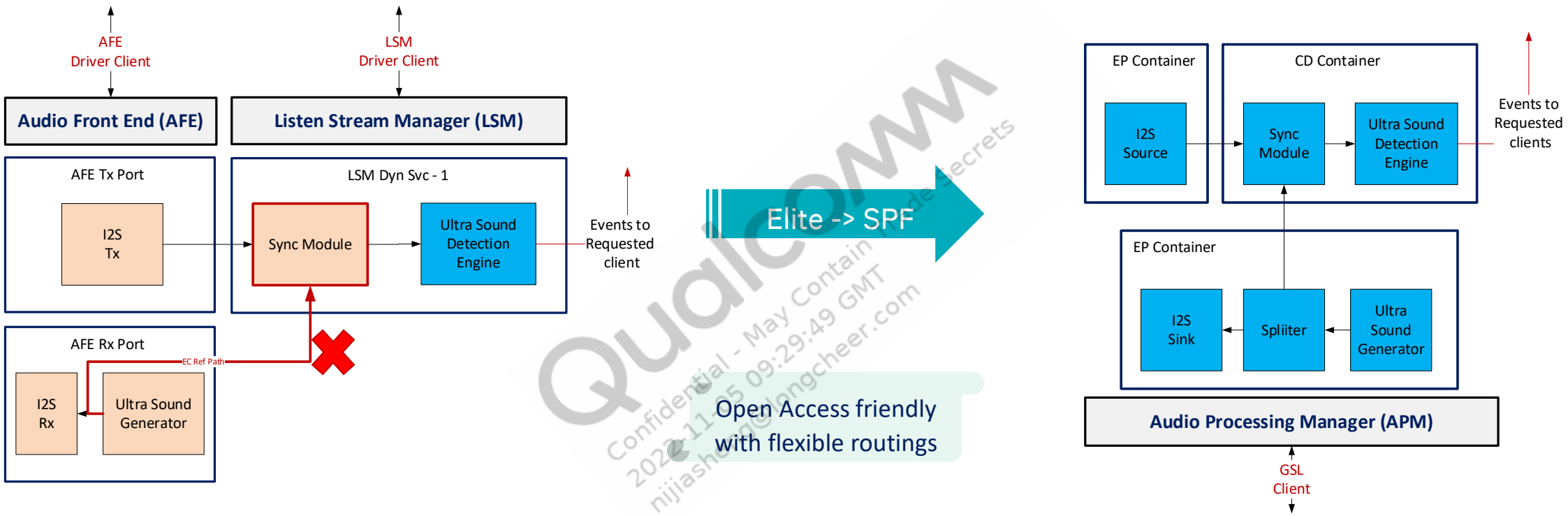


Optimal latencies with increased flexibility

Qualcomm Confidential – May Contain Trade Secrets
2022-11-09 09:29:49 GMT
nijiashenglongcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Ultrasound Proximity Detection



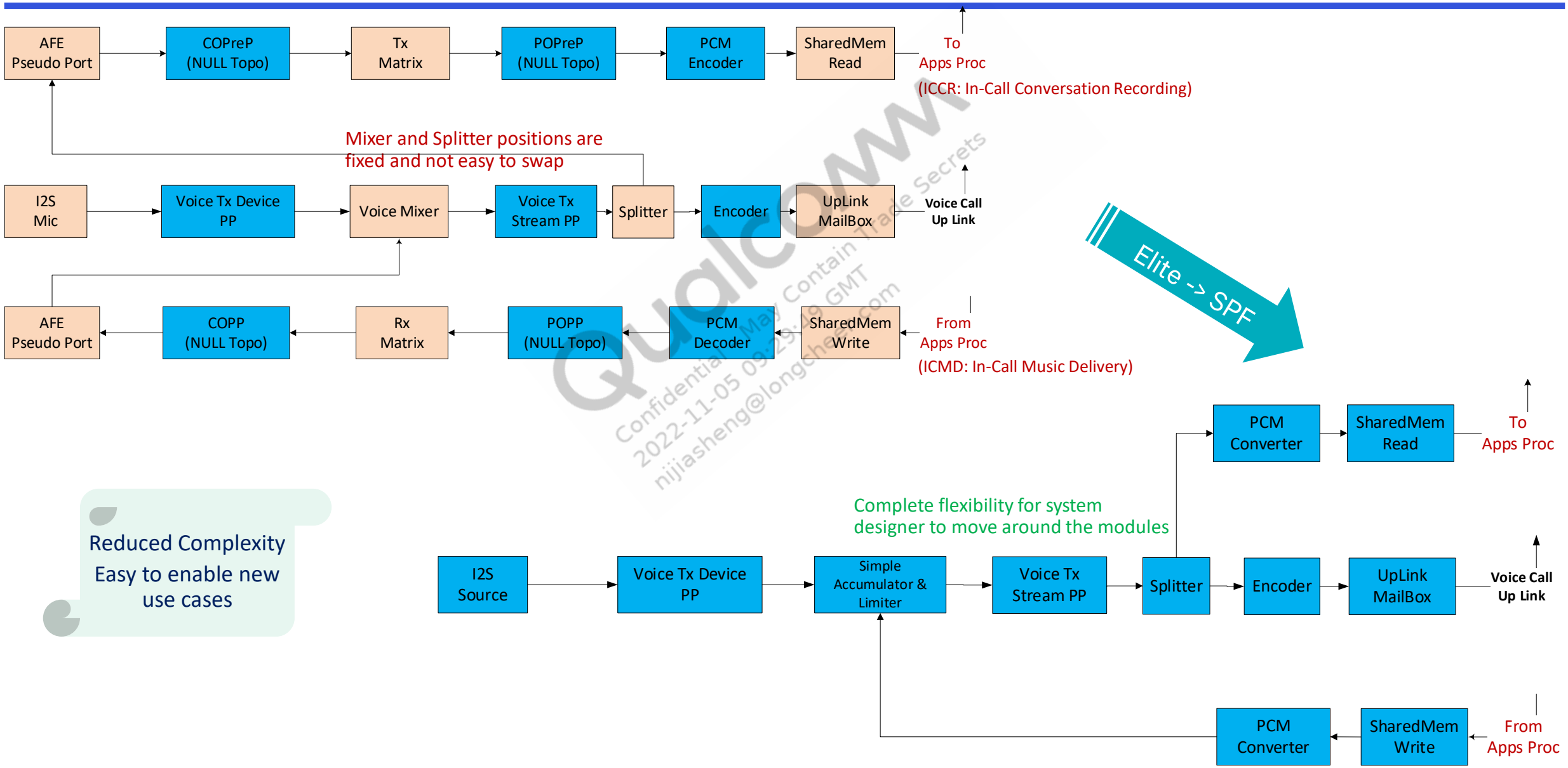
Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiashengcheer.com

- Rx path data routing to LSM not present
- Changes needed across the DSP and client drivers to enable

No changes are needed in SPF and the client driver to enable new use cases like ultrasound proximity detection

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

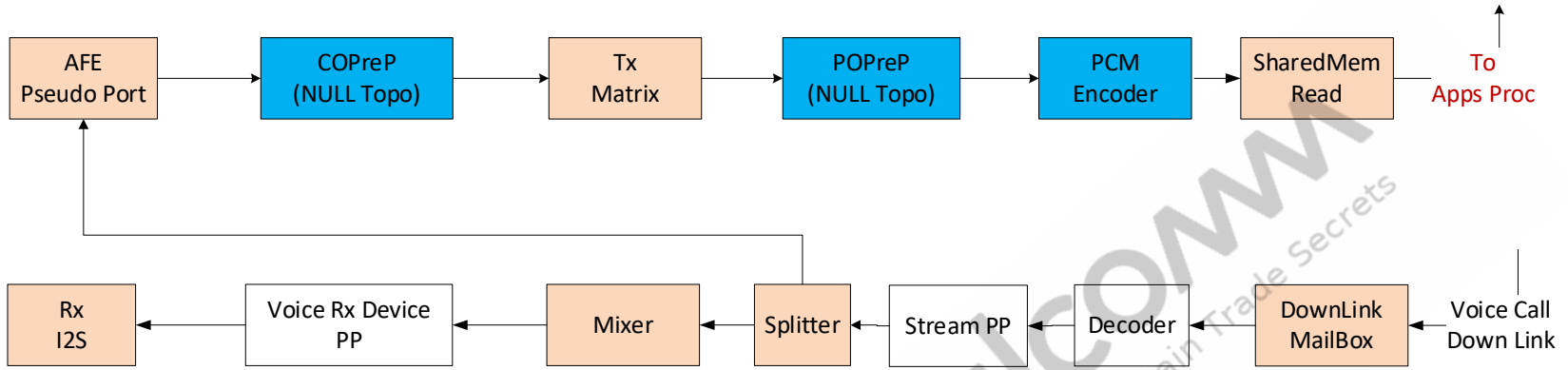
ICCR and ICMD Uplink



Reduced Complexity
Easy to enable new use cases

Qualcomm Confidential – May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longchee.com

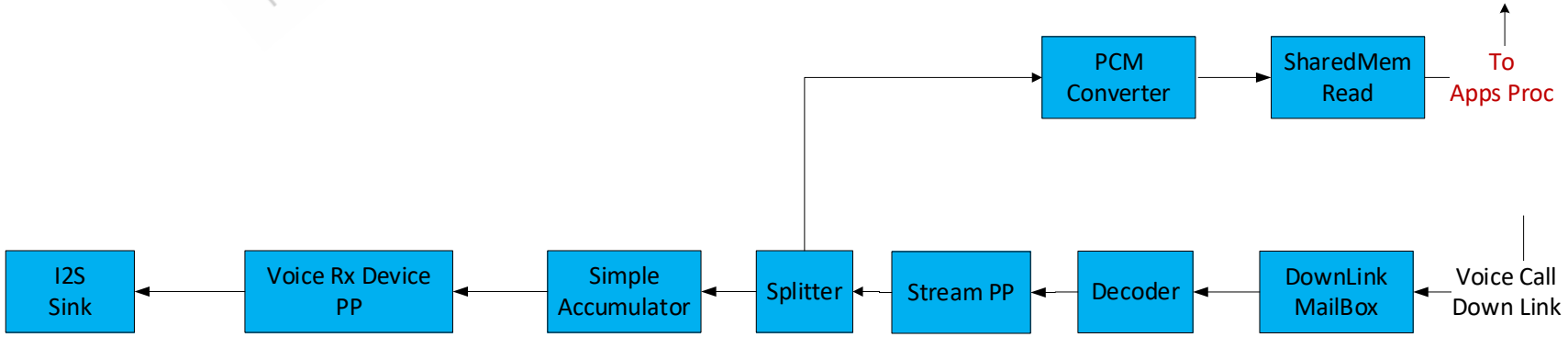
ICCR Downlink



Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

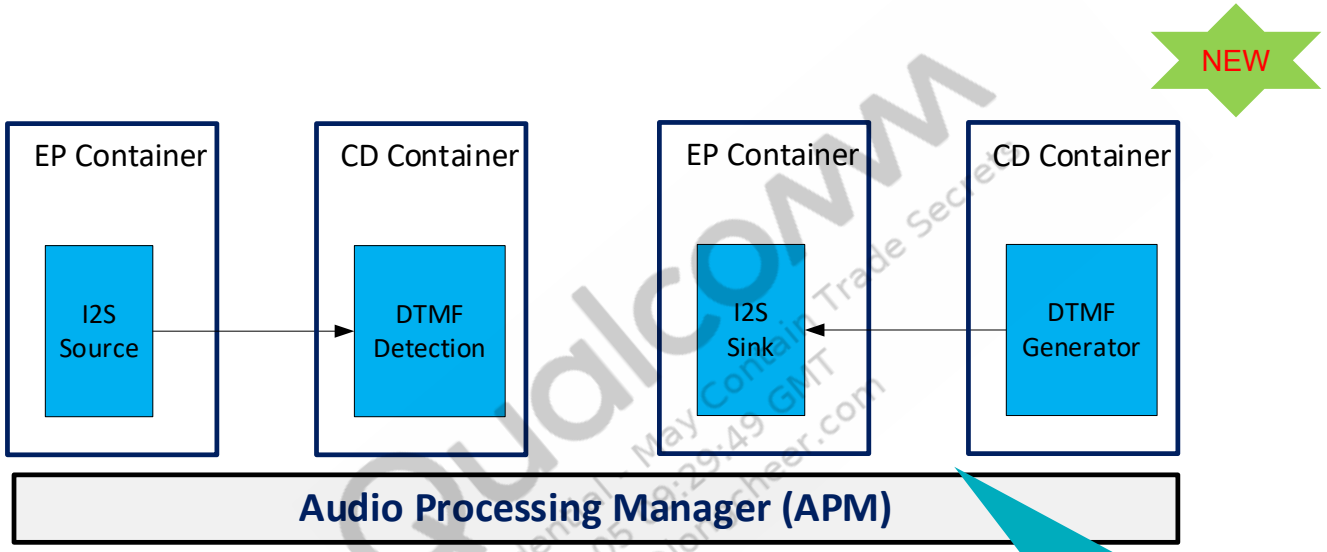
Elite ->SPF

Reduced Complexity
Easy to enable new use cases



Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

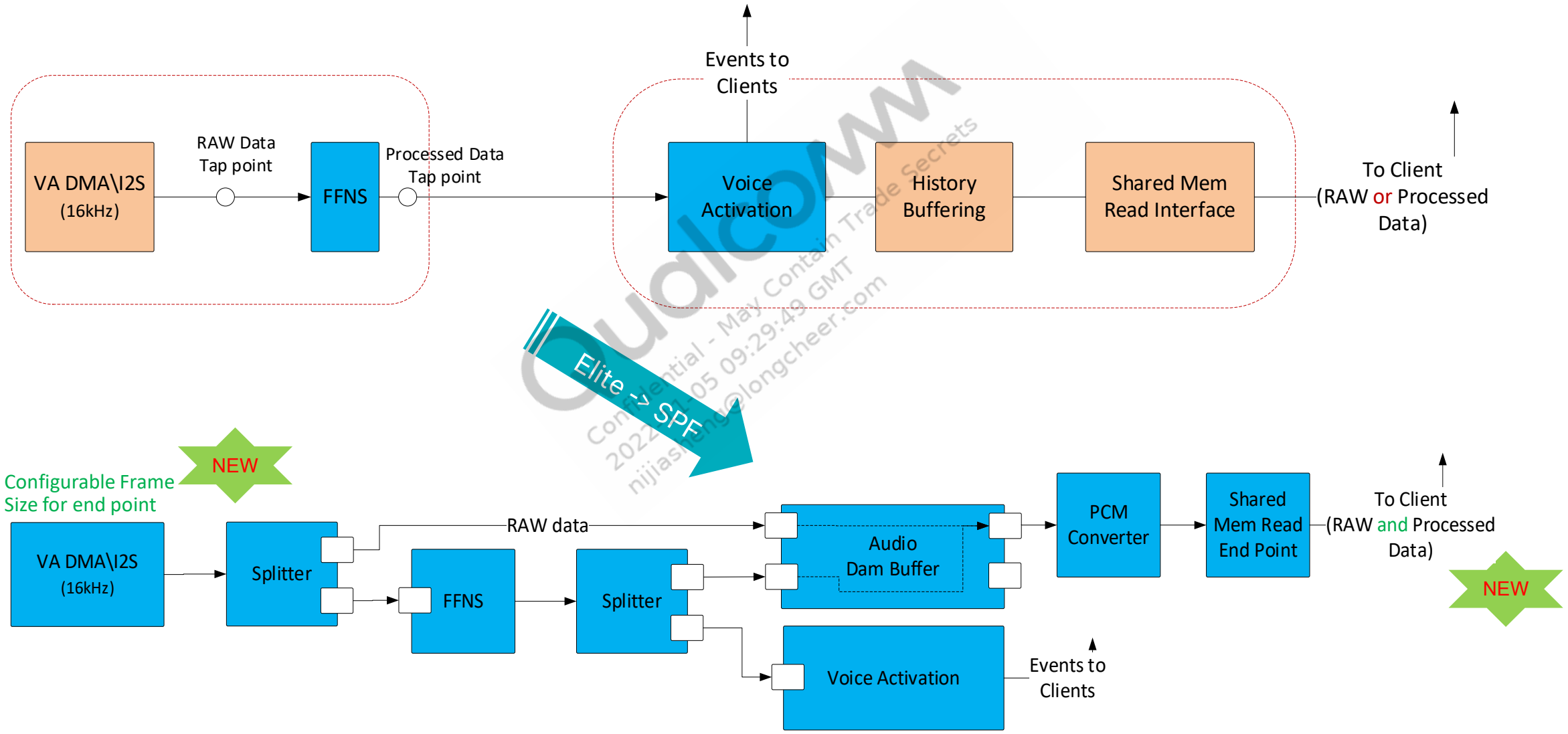
Source-Only and Sink-Only Modules



- Elite Framework did not support source-only or sink-only modules
- SPF supports these modules

Qualcomm Confidential – May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@lorscheer.com

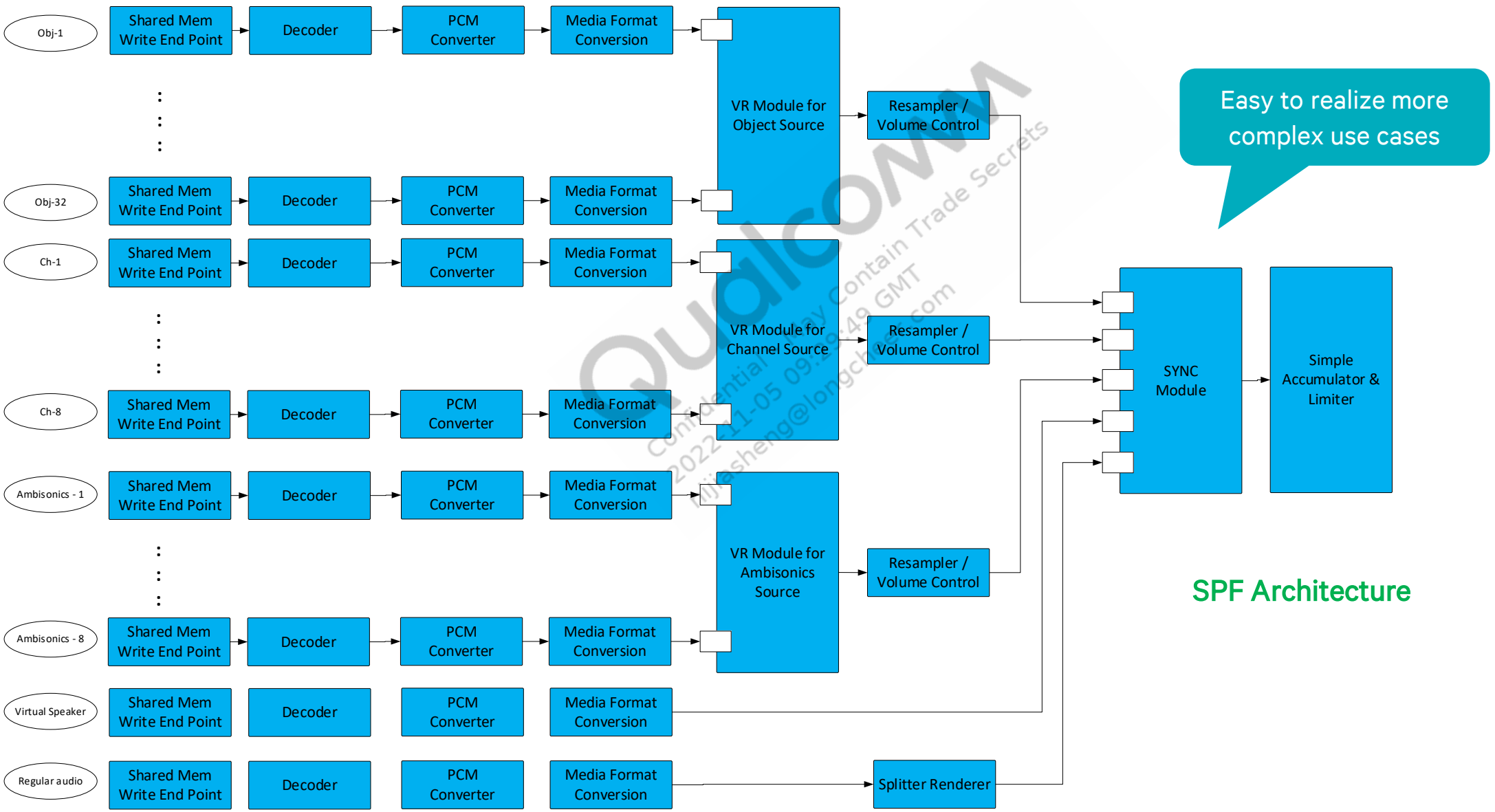
Voice UI



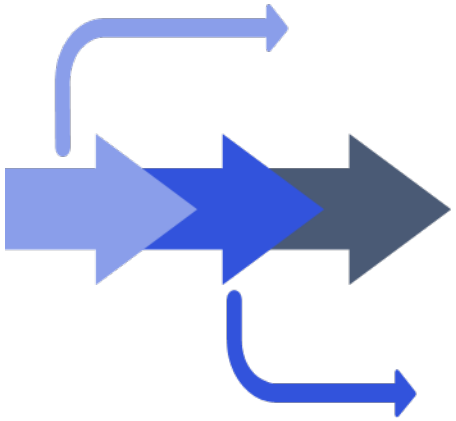
Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijiasheng@longcheer.com

Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

Virtual Reality (VR)



Qualcomm Confidential – May Contain Trade Secrets 2022-11-05 09:29:49 GMT mjesheng@longchee.com

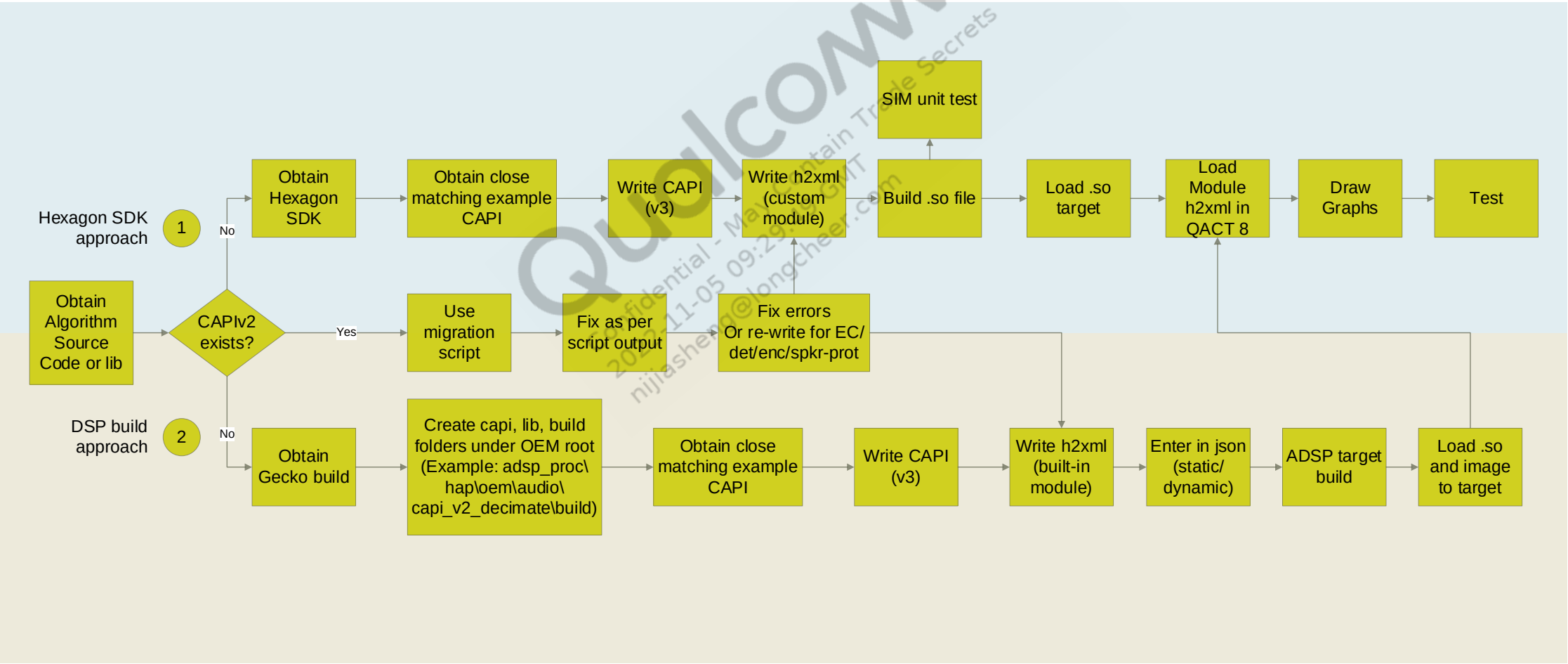


Workflow

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

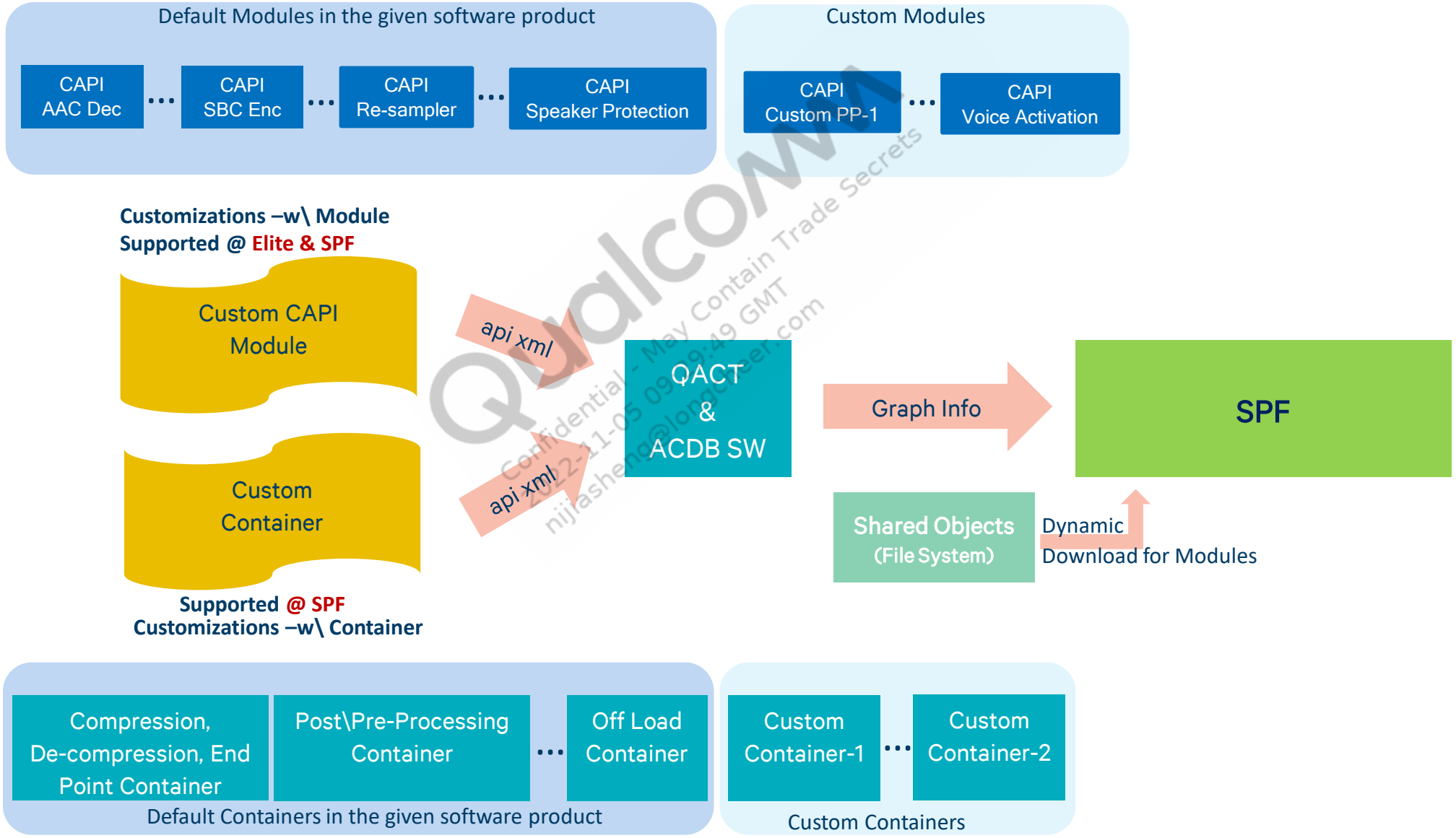
Module Integration Workflow

- See *Hexagon Access CAPIv2 Audio Custom Module Integration into Elite Framework from Command Line for AVS.ADSP.2.8/2.8.x* (80-NF772-43)



Confidential – Qualcomm Technologies, Inc. and/or its affiliated companies – May Contain Trade Secrets | MAY CONTAIN U.S. AND INTERNATIONAL EXPORT CONTROLLED INFORMATION

End to End Workflow



Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:09:49 GMT nijiasheng@lonestar.com

References

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com

References

Documents	
Title	**Number**
Qualcomm Technologies, Inc.	
Hexagon Access CAPIv2 Audio Custom Module Integration into Elite Framework from Command Line for AVS.ADSP.2.8/2.8.x	80-NF772-43

Acronyms	
Acronym or term	**Definition**
AMDB	Audio module database
DE	Drift estimators
GPR	SPF packet router
MID	Module ID
MIID	Module instance ID
OLC	Offload container
RM	Rate matching
RTC	Real-time calibration
RTM	Real-time monitoring
RTGM	Real time graph management

Qualcomm Confidential - May Contain Trade Secrets 2022-11-05 09:29:49 GMT nijasheng@longcheer.com

Questions?

https://createpoint.qti.qualcomm.com

Qualcomm
Confidential - May Contain Trade Secrets
2022-11-05 09:29:49 GMT
nijiasheng@longcheer.com